

# PAPER\_774: Tarantula Nebula 30 Doradus — UQFF Extreme Starburst LMC Evolution

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**Framework:** UQFF (Universal Quantum Field Superconductive Framework)  
**Session:** 181 | v5.41  
**Date:** 2026  
**CP4 Class:** #358 — TarantulaNebula30DorUQFFCalculator

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## Abstract

The Tarantula Nebula (30 Doradus, NGC 2070) in the Large Magellanic Cloud (~161,000 ly) is the most luminous HII region in the Local Group. Containing ~10<sup>5</sup> M<sub>☉</sub> of young stars including R136 (the densest known star cluster), it drives the most extreme stellar feedback observed in the nearby universe. Hubble’s mosaic shows spectacular filaments, pillars, and bow shocks spanning ~300 ly. Under UQFF, the extreme starburst magnetic field ( $B \approx 10^{-4}$  T), interaction velocity ( $v = 10^6$  m/s), and star-formation dynamics yield  $g_{\text{Tarantula}} \approx 1.053 \times 10^{-1}$  m/s<sup>2</sup> — the same class as major galaxy mergers, demonstrating UQFF’s convergence at extreme starburst conditions.

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## 1. Introduction

The Tarantula Nebula spans 1,000 ly in the LMC and would subtend 60° on the sky if placed at Orion’s distance. R136 alone contains hundreds of massive stars with total luminosity ~10<sup>7</sup> L<sub>☉</sub>, including several stars over 200 M<sub>☉</sub>. The feedback from O-type stars and Wolf-Rayet stars drives strong turbulence and amplifies the magnetic field to ~10<sup>-4</sup> T — 10× typical HII regions. Under UQFF, this B-field enhancement (starburst-induced), combined with the 10<sup>6</sup> m/s Aether coupling velocity, produces the same dominant term as the Antennae and Mice galaxy mergers, confirming that UQFF captures extreme starburst physics universally.

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## 2. Master UQFF Gravity Equation

$$g_{\text{Tarantula}}(r, t) = (G \times M) / r^2 \times (1 + H(z) \times t) \times (1 + M_{\text{sf}}) \times (1 - E_{\text{rad}}) \times (1 + f_{\text{TRZ}}) + a_{\text{EM}}$$

### 2.1 Parameters

| Parameter   | Symbol | Value                                                      | Source |
|-------------|--------|------------------------------------------------------------|--------|
| Nebula mass | M      | 10 <sup>5</sup> M <sub>☉</sub> = 1.989×10 <sup>35</sup> kg | Hubble |

| Parameter        | Symbol | Value                                                       | Source           |
|------------------|--------|-------------------------------------------------------------|------------------|
| Nebula radius    | r      | $3 \times 10^{17}$ m ( $\sim 31.7$ ly)                      | Hubble           |
| SFR              | SFR    | $5 \text{ M}_{\odot}/\text{yr}$                             | Labs             |
| Integration time | t      | $3 \times 10^6 \text{ yr} = 9.468 \times 10^{13} \text{ s}$ | Cluster age      |
| M_sf             | —      | 0.15                                                        | UQFF bound       |
| E_rad            | —      | 0.20                                                        | Extreme UV loss  |
| Redshift         | z      | 0.0005 (LMC)                                                | Distance         |
| v_EM             | v      | $10^6 \text{ m/s}$                                          | Starburst driven |
| B_starburst      | B      | $10^{-4} \text{ T}$                                         | Enhanced field   |
| f_TRZ            | —      | 0.1                                                         | UQFF             |

### 3. Long-Form Derivation

#### Step 1: Base Gravitational Term

$$g_{\text{grav}} = (6.6743\text{e-}11 \times 1.989\text{e}35) / (3\text{e}17)^2 \\ = 1.328\text{e}25 / 9\text{e}34 = 1.475\text{e-}10 \text{ m/s}^2$$

#### Step 2: Star-Formation Mass Fraction

$$M_{\text{sf}} = \text{SFR} \times t / M_0 = 5 \times 3\text{e}6 / 1\text{e}5 = 150 \rightarrow \text{UQFF bounded: } M_{\text{sf}} = 0.15 \\ 1 + M_{\text{sf}} = 1.15$$

#### Step 3: Radiation Energy Loss (R136 UV feedback)

$$\text{Extreme massive star feedback: } E_{\text{rad}} = 0.20 \text{ (much higher than M42)} \\ 1 - E_{\text{rad}} = 0.80$$

#### Step 4: Cosmic Expansion Factor

$$H(z) \text{ with } z = 0.0005 \text{ (LMC):} \\ H(z) = 2.268\text{e-}18 \times \sqrt{(0.3 \times (1.0005)^3 + 0.7)} \approx 2.268\text{e-}18 \text{ s}^{-1} \\ H(z) \times t = 2.268\text{e-}18 \times 9.468\text{e}13 = 2.147\text{e-}4 \\ 1 + H(z) \times t = 1.0002147$$

#### Step 5: Aether Electromagnetic Correction (Starburst Enhanced)

$$\text{Starburst feedback amplifies } B \text{ to } 10^{-4} \text{ T (10}\times \text{ normal HII)} \\ v = 10^6 \text{ m/s (turbulent starburst velocity)}$$

$$q \times (v \times B) = 1.602\text{e-}19 \times 1\text{e}6 \times 1\text{e-}4 = 1.602\text{e-}17 \text{ N} \\ a = 1.602\text{e-}17 / m_p = 1.602\text{e-}17 / 1.673\text{e-}27 = 9.575\text{e}9 \text{ m/s}^2 \\ a_{\text{EM}} = 9.575\text{e}9 \times 11 \times 1\text{e-}12 = 1.053\text{e-}1 \text{ m/s}^2$$

## Step 6: Time-Reversal Correction

$$1 + f_{\text{TRZ}} = 1.1$$

## Step 7: Final Solution

$$\begin{aligned} g_{\text{Tarantula}} &= (1.475\text{e-}10) \times (1.0002147) \times (1.15) \times (0.80) \times (1.1) + 1.053\text{e-}1 \\ &= 1.475\text{e-}10 \times 1.0002 = 1.475\text{e-}10 \\ &\times 1.15 = 1.696\text{e-}10 \\ &\times 0.80 = 1.357\text{e-}10 \\ &\times 1.1 = 1.493\text{e-}10 \\ &= 1.493\text{e-}10 + 1.053\text{e-}1 \\ &\approx 1.053\text{e-}1 \text{ m/s}^2 \end{aligned}$$

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## 4. Physical Interpretation

The Tarantula Nebula achieves the same UQFF result ( $1.053 \times 10^{-1} \text{ m/s}^2$ ) as galaxy-scale starbursts (M82, Antennae, Mice). This convergence is not coincidental — UQFF demonstrates that starburst-enhanced B-fields ( $10^{-4} \text{ T}$ ) universally produce this scaling regardless of whether the starburst is in a 300-ly LMC nebula or a 100-kly galaxy. The classical gravity contribution ( $1.493 \times 10^{-10} \text{ m/s}^2$ ) is negligible. R136's extreme stellar feedback is the Local Group's best analog for understanding cosmological starburst merger physics.

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## 5. UQFF Framework Advancement

- Confirms starburst  $B = 10^{-4} \text{ T}$  as universal starburst threshold across all scales
  - Tarantula = Local Group representative for galaxy-scale merger physics
  - UQFF unifies nebular and galactic starburst at  $1.053 \times 10^{-1} \text{ m/s}^2$  universal limit
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## 6. Conclusions

UQFF applied to the Tarantula Nebula (30 Doradus) yields  $g_{\text{Tarantula}} \approx 1.053 \times 10^{-1} \text{ m/s}^2$ , identical to galaxy merger starbursts. The starburst-amplified B-field ( $10^{-4} \text{ T}$ ) and turbulent velocity ( $10^6 \text{ m/s}$ ) combine to produce the universal extreme-starburst UQFF constant. This paper confirms that UQFF's starburst class ( $B = 10^{-4} \text{ T}$ ) applies from  $\sim 300$ -ly nebulae (Tarantula) to  $\sim 100$ -kly colliding galaxies (Antennae, Mice), establishing a scale-invariant starburst law.

*PAPER\_774, CP4 class #358. v5.41.*